

Computer Organization and Architecture		
Lab #:	10	
Lab Title:	Introduction to Verilog HDL Combinational Circuits 1	
Submitted by:		
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Rubrics name & number					Marks	
					In-Lab	Post-Lab
Engineering Knowledge	R2: Use of Engineering Knowledge and follow Experiment Procedures: Ability to follow experimental procedures, control variables, and record procedural steps on lab report.					
Problem Analysis	R6: Experimental Data Analysis: Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.					
Design	R8: Best Coding Standards: Ability to follow the coding standards and programming practices.					
Modern Tools Usage	R9: Understand Tools: Ability to describe and explain the principles behind and applicability of engineering tools.					
Individual and Teamwork	R12: Individual Work Contributions: Ability to carry out individual responsibilities.					
	R13: Management of Team Work: Ability to appreciate, understand and work with multidisciplinary team members.					
Rubrics #	R2	R6	R8	R9	R12	R13
In Lab						
Post- Lab						

LAB: 10

Tasks (Use Xilinx):

- Write the Verilog gate-level description and testbench for a circuit with 6-bit input(X) and 1-bit output (Y). It compares the first three bits of the inputs with the latter three and sets output equal to 1 when both are equal.

$$Y = X_0 * X_3 + X_1 * X_4 + X_2 * X_5.$$

Verilog:

```
module compare_bits(  
input [5:0] X,    // Inputs of UUT decided as ?reg?  
output Y);       //Outputs of UUT (is this case module verilog?) decided as // wire  
wire X0, X1, X2, X3, X4, X5;  
wire AND1, AND2, AND3;  
wire OR1;  
  
assign X0 = X[0];  
assign X1 = X[1];  
assign X2 = X[2];  
assign X3 = X[3];  
assign X4 = X[4];  
assign X5 = X[5];  
  
AND gate1(AND1, X0, X3);  
AND gate2(AND2, X1, X4);  
AND gate3(AND3, X2, X5);  
OR gate4(OR1, AND1, AND2);  
OR gate5(Y, OR1, AND3);
```

```
endmodule
```

TestBench:

```
module compare_bits(  
input [5:0] X, // Inputs of UUT decided as ?reg?  
output Y); //Outputs of UUT (is this case module verilog?) decided as // wire  
wire X0, X1, X2, X3, X4, X5;  
wire AND1, AND2, AND3;  
wire OR1;  
  
// Assign individual bits of X to wires  
assign X0 = X[0];  
assign X1 = X[1];  
assign X2 = X[2];  
assign X3 = X[3];  
assign X4 = X[4];  
assign X5 = X[5];  
  
// AND gates  
and (AND1, X0, X3); // AND1 = X0 AND X3  
and (AND2, X1, X4); // AND2 = X1 AND X4  
and (AND3, X2, X5); // AND3 = X2 AND X5  
  
// OR gates  
or (OR1, AND1, AND2); // OR1 = AND1 OR AND2  
or (Y, OR1, AND3); // Y = OR1 OR AND3  
  
endmodule
```

Result:

